

CHETHIYA RATHNASEKARA

Full-Stack Software Engineering Undergraduate

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SUMMARY

Full-stack software engineering undergraduate at Java Institute (since 2023) who builds desktop, web and AI applications in Java (Swing, JSP/Servlets), React/Next.js, Node.js and Python. Built and released 8+ personal projects, including a real-time MetaTrader 5 analytics app (under 100 ms sync) and a Java POS system. Oracle and Aviatrix certified; looking for a software engineering internship.

PROJECTS

Traders Journal (KING Edition v1.0.8)

2026 GitHub Release

Electron 41, Node.js, Python, MQL5, SQLite/JSON, NSIS

- Built a privacy-first Windows trading analytics app with a custom MQL5 Expert Advisor and a localhost HTTP/IPC bridge to Electron, syncing live MT5 deals in under 100 ms with no manual setup.
- Implemented equity curves, a daily PnL calendar and multi-account login; calculates 15+ risk metrics (win rate, profit factor, max drawdown) fully offline.
- Packaged it with NSIS and PyInstaller, bundling a portable Python runtime, cutting setup from 30+ minutes of manual configuration to under 60 seconds.

POS & Inventory Management System

2026 GitHub Release

Java, Swing, FlatLaf, JasperReports, MySQL 8

- Developed a desktop retail POS with role-based access (Admin/Cashier) and JasperReports receipts and reports; barcode checkout and stock deduction complete in under 3 seconds per sale.

Online Bookstore Web Application

Java EE (JSP/Servlets), JDBC, MySQL, Bootstrap, JavaScript

- Engineered a full-stack bookstore with authentication, catalog filtering and cart checkout on a normalized MySQL schema (JDBC), plus an RBAC admin dashboard for inventory, pricing and orders.

Orbit Media Studio (Galaxy PWA)

2026 GitHub

Next.js 15, TypeScript, Node.js, Tailwind CSS, FFmpeg, Docker

- Built an installable PWA with a Node.js media engine (link detection across 11+ video platforms, batch queues) and a Dockerized FFmpeg pipeline for batch watermark removal.

ScreenToChatGPT & Daybook

2026 Screen-Capture Daybook

Python, PyWin32, GPT-4o API, Electron, React 18, TypeScript

- Built a global-hotkey multi-monitor capture tool that sends screenshots to ChatGPT Desktop in 240 ms, and Daybook, an Electron/React trading journal with live ForexFactory news.

4K AI Video Upscaler

Python, PyTorch (CUDA), Real-ESRGAN, FFmpeg, Next.js

- Built a CUDA-accelerated Real-ESRGAN 4K 60 FPS video super-resolution pipeline with multi-threaded FFmpeg processing, raising batch rendering throughput about 3x.

Other Projects

- **Enterprise Academy & Inventory System** (Java Swing, MigLayout, MySQL): registration, scheduling and billing with automated PDF invoices and reports.
- **Cloud-Backed Web App & Portfolio** (React, Tailwind, Supabase, Netlify): Supabase auth and real-time PostgreSQL sync, deployed via Netlify CI/CD.

SKILLS

Frontend

React 18 | Next.js 15 | TypeScript | JavaScript | HTML5 | CSS3 | Tailwind CSS | Bootstrap | PWA

Backend & Desktop

Java | Swing | JSP/Servlets | JDBC | JasperReports | Node.js | Express.js | REST APIs | Python 3 | Electron | MQL5

Databases

MySQL 8 | PostgreSQL | Supabase | SQLite | Oracle DB | Oracle APEX

DevOps & Cloud

Git | Docker | GitHub Actions | Netlify | NSIS | PyInstaller | Oracle Cloud (OCI) | Aviatrix

AI, Media & Testing

GPT-4o API | PyTorch CUDA | Real-ESRGAN | FFmpeg | Postman | Unit & Integration Testing

EDUCATION

Undergraduate, Software Engineering (BEng/BSc)
Java Institute for Advanced Technology
2023 – Present Sri Lanka

G.C.E. Advanced Level – Technology Stream
2025 (results 2026)

G.C.E. Ordinary Level
2023 (results 2024)

CERTIFICATIONS

Oracle APEX Cloud Certified Developer Professional · Oracle 2025

OCI Foundations Associate · Oracle 2025

OCI AI Foundations Associate · Oracle 2025

OCI Data Platform Foundations Associate · Oracle 2025

Aviatrix Certified Engineer (ACE), Multi-Cloud Network Associate · Aviatrix 2025

HACKATHONS

- 24-Hour Software Application Development Hackathon (2025)
- Robotics Hackathon
- IoT Hackathon

LANGUAGES & DETAILS

Sinhala – Native
English – Professional working proficiency
Date of birth – 22 April 2006